

Dual-Domain MMD Adaptation: Regularization Ablation in the Large-Source, Small-Target Regime

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1 Setup

All five variants below share the same starting checkpoint (`syn_large` pretrain baseline) and the same target set (`real_small`), differing only in whether/how MMD regularization was applied:

- **`real_small_from_syn_large`** (no-MMD control): plain continued fine-tune, no MMD term, no source batches. 100 epochs, batch 8.
- **`syn_large_source_real_small_target`** (unregularized MMD): constant `mmd.weight = 0.8` (default), no bandwidth freeze. 50 epochs, batch 8. See `experiment_summary_v4.tex`.
- **`syn_large_source_real_small_target_reg`**: `mmd.weight` decaying linearly $1.0 \rightarrow 0.1$, bandwidth freeze at epoch 1. 50 epochs.
- **`_reg_smaller_weight`**: `mmd.weight` decaying linearly $0.9 \rightarrow 0.02$, bandwidth freeze at epoch 1. 50 epochs, batch 4.
- **`_reg_smaller_weight_longer_run_final`**: same weight schedule as `_reg_smaller_weight` ($0.9 \rightarrow 0.02$, freeze at epoch 1), but 100 epochs instead of 50.

A note on hyperparameter provenance. None of the MMD-specific settings (weight, schedule, freeze epoch) are captured in ultralytics’ own `args.yaml` – they are popped from the overrides dict before being handed to the stock trainer, so they leave no automatic record. The values above for the three regularized runs are as recalled after the fact rather than read from a log; treat the base `_reg` run’s freeze epoch in particular as approximate.

2 Results

3 Discussion

Regularization recovers a small, consistent MMD benefit on the target domain. In `experiment_summary_v4.tex`, unregularized MMD showed no visible advantage over the no-MMD control on vehicle/real (0.877 vs. 0.877 mAP50, identical). Here, all three regularized variants edge out both baselines on this exact metric: 0.881, 0.879, and 0.882 vs. 0.877 for the two baselines. This is a small margin (0.2–0.6 points), but it is now three independent confirmations pointing the same direction rather than one ambiguous run – a materially stronger signal than the unregularized

Model	p mAP50	p mAP50-95	v mAP50	v mAP50-95
real_small_from_syn_large (no-MMD)	0.644	0.303	0.877	0.626
syn_large_source_real_small_target (unreg. MMD)	0.642	0.296	0.877	0.625
_reg	0.649	0.300	0.881	0.628
_reg_smaller_weight	0.646	0.302	0.879	0.625
_reg_smaller_weight_longer_run_final	0.651	0.304	0.882	0.630

Table 1: Validated on the real val set – the target domain for this direction, the metric of primary interest.

Model	p mAP50	p mAP50-95	v mAP50	v mAP50-95
real_small_from_syn_large (no-MMD)	0.129	0.0349	0.222	0.146
syn_large_source_real_small_target (unreg. MMD)	0.111	0.0266	0.214	0.138
_reg	0.103	0.0255	0.218	0.141
_reg_smaller_weight	0.116	0.0282	0.227	0.145
_reg_smaller_weight_longer_run_final	0.0886	0.0216	0.215	0.140

Table 2: Validated on the syn val set – the source (forgotten) domain for this direction.

case gave. The best performer overall is `_reg_smaller_weight_longer_run_final` (100 epochs, smaller decaying weight), which leads on both vehicle *and* person, on both val sets simultaneously.

The source (forgetting) domain shows no consistent pattern. On syn, `_reg_smaller_weight` has the best vehicle score of all five variants (0.227), but `_reg_smaller_weight_longer_run_final` – the best performer on the target domain – has one of the weaker person scores on syn (0.0886, worse than every other variant). Regularization does not appear to uniformly help forgetting the way it helps the target-domain result; the effect there is mixed rather than directional.

Caveats. Margins throughout are small (a few tenths of a point), these are single runs with no repeated seeds, and the exact hyperparameters for the base `_reg` run are recalled rather than logged (see note in Setup). The directional consistency across three separately-configured regularized runs is encouraging, but confirming this with a repeat run or a small sweep around the smaller-weight/longer-run setting would be the natural next step before treating the magnitude of the effect as settled.